

Agentic Customer 360 – Mid-Term Approach Report

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PS: Agentic Customer 360 – Proactive Intervention Desk

1. Problem and Goal

The PS requires an ambient Customer 360 system, not a passive dashboard or chatbot. The system must continuously consume asynchronous multi-source events, maintain evolving customer state, infer relevant life phases, coordinate specialized agents, and commit to an explicit action or No Action.

2. What the Research Taught Us

Streaming: CEP and rolling-window processing should reduce noisy raw events before expensive agentic reasoning.

Memory: Working, episodic and semantic memory have different roles; historical evidence should be retained while its retrieval influence can decay.

Multi-agent coordination: Swarm is suitable for independent signal collection, while handoff, debate and critique-refiner fit later decision stages.

Governance: Deterministic guardrails, scoped access, bounded tools, provenance and real HITL are necessary when decisions can affect customers or money.

3. Proposed Architecture

Event stream → PII protection → deterministic guardrail → CEP / rolling features → per-customer State Board → tripartite memory → specialist swarm → life-event inference → synthesis → debate when needed → bounded action proposal → critique / compliance → HITL → terminal action → `inferred-events.jsonl` + trace.

4. Key Architectural Decisions

Decision	Reason
Python-native CEP-lite	Preserves streaming semantics while keeping the prototype buildable.
Governed tripartite memory	Separates short-term context, historical evidence and shared policy knowledge.
Decay-weighted episodic retrieval	Addresses stale historical signals without deleting ground-truth history.
Swarm + Debate + Judge	Matches coordination topology to independent analysis and genuine conflicts.
Critique-Refiner with max 2 passes	Provides a dedicated policy/cost/tone review before human approval.
Deterministic guardrails + RBAC + bounded tools	Prevents the LLM from bypassing hard safety and access constraints.
Real HITL interruption	Allows approve / modify / reject and escalation for ambiguity, not only cost.

5. Expected Decision Flow

Specialist agents publish structured findings to the State Board. The Life-Event Agent maintains an evolving belief state. Synthesis determines whether evidence supports No Action or requires arbitration. Genuine conflicts go to Debate + Judge. A bounded action proposal is then critiqued before a real HITL approval checkpoint.

6. Intended Contribution

The individual components are established in prior research and existing systems. The proposed contribution is their deliberate composition into a governed ambient Customer 360 workflow that combines streaming, customer-scoped memory, life-event state, stage-specific multi-agent coordination, bounded actions, HITL and traceable outputs.

7. Main Tradeoffs

The design favors prototype buildability over production infrastructure. Debate and critique add latency, so they are conditional. Life-event inference remains probabilistic, and sensitive or low-confidence cases are routed to human review.